KS3 DIGITAL TECHNOLOGY

VR, AR AND MIXED REALITY

Stepping into digital worlds: the difference between virtual, augmented and mixed reality, how headsets trick your brain into believing, how AR apps understand the real world, how XR is used in training, health and education, how to use it safely and ethically, and how to design your own XR experience.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 The reality spectrum
2 Tricking the brain
3 AR in your pocket
4 XR at work and school
5 Safe and ethical XR
6 Design an XR experience

LESSON 1 OF 6

THE REALITY SPECTRUM

Virtual reality (VR) replaces the real world with a digital one.

STARTER · 5 MINUTES

DO NOW

What's the difference between watching a film, playing a video game and wearing a VR headset?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Define VR, AR, mixed reality and XR and give examples of each

SUCCESS CRITERIA: I CAN...

- ✓ I can define VR, AR and MR
- ✓ I can place XR experiences on the reality spectrum
- ✓ I can give examples of XR devices and apps

THREE REALITIES

- VR: replaces your surroundings with a digital world
- AR: adds digital information on top of the real world
- MR: digital objects that react to and interact with the real world
- XR (extended reality): the umbrella term for all of them

KEY WORDS

VIRTUAL REALITY	Technology that replaces the real world with a computer-generated one.
AUGMENTED REALITY	Technology that adds digital content on top of the real world.
MIXED REALITY	Technology where digital objects understand and interact with the real world.
EXTENDED REALITY	The umbrella term for VR, AR and MR.
IMMERSION	How completely an experience surrounds your senses.

THE REALITY SPECTRUM

- Real world → AR → MR → VR → fully virtual
- Phone AR: Snapchat filters, Pokémon GO, IKEA Place
- Mixed reality headsets: Meta Quest 3, Apple Vision Pro
- VR: games, virtual field trips, simulations

ACTIVITY · 15 MINUTES · PAIRS

REALITY SORT

1. Sort the 12 example cards (e.g. a Pokémon GO battle, a VR rollercoaster, a sat-nav arrow on a car windscreen, a virtual pet that hides behind your real sofa) into VR, AR and MR.
2. Place them on a reality spectrum line from "real" to "virtual".
3. Choose one and suggest how it could move further along the spectrum.

In your workbook, answer:

? Where does a virtual pet hiding behind your real sofa go, and why?

? How could a phone AR game become more like mixed reality?

Your workbook: Lesson 1 · tick each step off as you go

XR DEVICES

- Phones and tablets: the most common AR devices
- Standalone headsets: everything built in, no PC needed
- Passthrough: cameras show you the real room inside a headset
- Smart glasses: lightweight; cameras, audio and small displays

QUICK CHECK

1. What does VR do?

A Replaces the real world with a digital one
B Adds labels to the real world
C Records video
D Prints 3D objects

2. What does AR do?

A Turns off your camera
B Adds digital content on top of the real world
C Plays audio only
D Replaces the real world completely

3. What is XR?

A A new type of X-ray
B A phone brand
C The umbrella term for VR, AR and MR
D A game console

4. Which is an example of AR?

A A radio
B A face filter on a social media app
C A VR rollercoaster
D A paper map

QUICK CHECK: ANSWERS

1. **A: Replaces the real world with a digital one**
   Fully virtual.

2. **B: Adds digital content on top of the real world**
   Augments reality.

3. **C: The umbrella term for VR, AR and MR**
   Extended reality.

4. **B: A face filter on a social media app**
   Digital content on the real world.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between VR, AR and MR.

Look back at today's success criteria: can you tick them all?

- ✓ I can define VR, AR and MR
- ✓ I can place XR experiences on the reality spectrum
- ✓ I can give examples of XR devices and apps

LESSON 2 OF 6

TRICKING THE BRAIN

Headsets show a slightly different image to each eye so your brain sees depth (stereoscopic 3D).

STARTER · 5 MINUTES

DO NOW

Hold a finger in front of your face and close one eye, then the other. What happens?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how XR headsets use displays, lenses, tracking and low latency to create presence, and why motion sickness happens

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how stereoscopic displays create depth
- ✓ I can explain tracking and six degrees of freedom
- ✓ I can explain presence, latency and motion sickness

SEEING IN 3D

- Your eyes see slightly different views; your brain combines them to judge depth
- Headsets show a different image to each eye: stereoscopic display
- Lenses focus the screens and stretch them to fill your field of view
- Higher resolution reduces the "screen door" effect

KEY WORDS

STEREOSCOPIC	Showing a slightly different image to each eye to create a 3D effect.
FIELD OF VIEW	The area you can see without moving your eyes or head.
SIX DEGREES OF FREEDOM	Tracking both rotation and movement up/down, left/right and forward/back.
LATENCY	The delay between an action and the response, such as moving your head and the image changing.
PRESENCE	The feeling of really being in a virtual world.

TRACKING

- 3DoF: tracks head turning only
- 6DoF: tracks turning AND moving: up/down, left/right, forward/back
- Inside-out tracking: cameras on the headset map the room
- Hand and eye tracking let you interact without controllers

ACTIVITY · 20 MINUTES · PAIRS

DEPTH AND DELAY EXPERIMENTS

1. Try the finger test: hold a pencil at arm's length and try to touch its tip with another pencil using one eye, then both eyes. Record your success rate.
2. Use a cardboard viewer or stereo image cards (if available) to see 3D.
3. Watch two video clips: one smooth and one with lag (your teacher will provide them). Describe how each feels.
4. Explain your results using stereoscopic, latency and presence.

In your workbook, answer:

? Why was it harder to touch the pencil with one eye?

? Why does lag make XR uncomfortable?

Your workbook: Lesson 2 · tick each step off as you go

PRESENCE AND COMFORT

- Presence: the feeling of really being there
- Needs high refresh rate (90 Hz or more) and low latency (under about 20 ms)
- If what you see doesn't match what your inner ear feels, you may feel sick
- Comfort settings: teleport movement, snap turning, vignettes

QUICK CHECK

1. What is a stereoscopic display?

A A black-and-white screen
B It shows a different image to each eye to create depth
C A curved TV
D A display with speakers

2. What does 6DoF tracking add compared with 3DoF?

A A bigger screen
B More colours
C Better sound
D Tracking movement, not just head rotation

3. What is presence in XR?

A A gift
B Being on time
C The feeling of really being there
D The headset's weight

4. Why does XR motion sickness happen?

A The headset is too cold
B Bright colours
C Your eyes and inner ear disagree about movement
D The battery is low

QUICK CHECK: ANSWERS

1. **B: It shows a different image to each eye to create depth**
   Two views, one 3D world.

2. **D: Tracking movement, not just head rotation**
   Step, crouch, lean.

3. **C: The feeling of really being there**
   Immersion's goal.

4. **C: Your eyes and inner ear disagree about movement**
   Sensory mismatch.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a headset creates a 3D image.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how stereoscopic displays create depth
- ✓ I can explain tracking and six degrees of freedom
- ✓ I can explain presence, latency and motion sickness

LESSON 3 OF 6

AR IN YOUR POCKET

AR apps use the camera, motion sensors and computer vision to understand the real world.

STARTER · 5 MINUTES

DO NOW

How does a phone know where your floor is, so a virtual sofa doesn't float in the air?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how AR apps detect and map the real world, and describe uses of AR

SUCCESS CRITERIA: I CAN...

- ✓ I can describe marker-based, markerless and location-based AR
- ✓ I can explain how SLAM lets a device map a space
- ✓ I can describe useful applications of AR

THREE TYPES OF AR

- Marker-based: recognises a picture or QR code and adds content to it
- Markerless: detects floors, tables and walls to place objects anywhere
- Location-based: uses GPS to show content at real places
- Face tracking: maps facial features for filters

KEY WORDS

MARKER-BASED AR	AR that recognises a specific image or code and adds content to it.
MARKERLESS AR	AR that detects surfaces and tracks movement so objects can be placed anywhere.
SLAM	Simultaneous Localisation And Mapping: building a map of a space while tracking your position in it.
PLANE DETECTION	Finding flat surfaces such as floors, tables and walls.
ANCHOR	A fixed point in an AR map that keeps a virtual object in place.

HOW AR SEES THE WORLD

- Camera images + motion sensors (accelerometer, gyroscope)
- SLAM: Simultaneous Localisation And Mapping
- Plane detection finds flat surfaces
- Anchors keep virtual objects fixed in place as you move

AR AT WORK AND PLAY

- Games: Pokémon GO, AR board games
- Shopping: try on glasses, place furniture
- Navigation: arrows in the street; car head-up displays
- Translation: point at a sign and read it in your language

ACTIVITY · 25 MINUTES · PAIRS

AR INVESTIGATION

1. Using a school tablet (or the demo videos), try an AR app that places objects (e.g. a measuring app or a 3D model viewer).

2. Test placing objects on different surfaces: carpet, desk, white wall, window. Record which work best.

3. Identify which type of AR the app uses.

4. Design an AR feature for your school (e.g. a tour for new Year 8 pupils).

In your workbook, answer:

? Which surfaces worked best and worst, and why?

? Describe your AR feature for the school.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What does marker-based AR recognise?

A Your heartbeat
B Wi-Fi signals
C Your voice
D A specific image or code

2. What does SLAM stand for?

A Super Large AR Module
B Simultaneous Localisation And Mapping
C Slow Loading AR Map
D Smart Light And Motion

3. What does plane detection find?

A Flat surfaces like floors and tables
B Aeroplanes
C Wi-Fi networks
D Faces

4. Why does AR struggle on plain white walls?

A AR only works outdoors
B Walls are too big
C White is too bright
D There are few distinctive features to track

QUICK CHECK: ANSWERS

1. **D: A specific image or code**
   Markers trigger content.

2. **B: Simultaneous Localisation And Mapping**
   Map and locate at once.

3. **A: Flat surfaces like floors and tables**
   Places to put objects.

4. **D: There are few distinctive features to track**
   Features are needed.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how a markerless AR app places a virtual object on a table.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe marker-based, markerless and location-based AR
- ✓ I can explain how SLAM lets a device map a space
- ✓ I can describe useful applications of AR

LESSON 4 OF 6

XR AT WORK AND SCHOOL

XR lets people practise dangerous, expensive or rare situations safely, from surgery to firefighting.

STARTER · 5 MINUTES

DO NOW

Which would you rather learn first in VR: flying a plane, performing surgery or defusing a bomb? Why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how XR is used in training, healthcare, education and industry, and evaluate its benefits and limitations

SUCCESS CRITERIA: I CAN...

- ✓ I can describe uses of XR in different industries
- ✓ I can explain why XR is useful for training
- ✓ I can evaluate the benefits and limitations of XR

TRAIN SAFELY

- Pilots have trained in flight simulators for decades
- Surgeons rehearse operations in VR
- Firefighters, police and soldiers practise emergencies
- Mistakes in a simulation hurt no one

KEY WORDS

SIMULATION	A computer model of a real situation used for testing or training.
VIRTUAL FIELD TRIP	Visiting a place using VR or other digital technology.
REMOTE ASSISTANCE	An expert helping someone from a distance by seeing what they see.
EXPOSURE THERAPY	Treatment that helps people overcome fears by facing them gradually.

LEARN AND HEAL

- Virtual field trips: the Moon, the Titanic, inside a cell
- Exposure therapy: facing fears like heights or spiders step by step
- Pain distraction during treatment
- Rehabilitation exercises turned into games

DESIGN AND WORK

- Architects walk clients through buildings before they're built
- Engineers inspect digital twins of factories and products
- Remote assistance: an expert sees what you see and draws instructions
- Virtual meetings with 3D models

ACTIVITY · 25 MINUTES · GROUPS OF 3

XR CONSULTANT

1. Your team is advising one organisation: a hospital, a fire service, a farm, or a secondary school.
2. Suggest two ways they could use XR.
3. Complete a benefits and limitations table for your suggestions.
4. Write a recommendation: should they invest in XR, and how much?

In your workbook, answer:

? Describe your two XR uses.

? Write your recommendation.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. Why is XR useful for training pilots and surgeons?

A It's always cheaper than everything
B It's more colourful
C It replaces all real training
D They can practise dangerous situations safely

2. What is a virtual field trip?

A A school trip by bus
B A sports match
C Visiting a place in VR
D A video call

3. What is remote assistance in AR?

A A remote control
B A robot helper
C An expert far away sees what you see and guides you
D A phone call only

4. Which is a limitation of XR in schools?

A Headsets are expensive
B It's memorable
C It's engaging
D It's safe

QUICK CHECK: ANSWERS

1. **D: They can practise dangerous situations safely**
   Safe, repeatable practice.

2. **C: Visiting a place in VR**
   Go anywhere.

3. **C: An expert far away sees what you see and guides you**
   Shared view plus instructions.

4. **A: Headsets are expensive**
   Cost and practicality.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why VR is useful for training firefighters.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe uses of XR in different industries
- ✓ I can explain why XR is useful for training
- ✓ I can evaluate the benefits and limitations of XR

LESSON 5 OF 6

SAFE AND ETHICAL XR

XR can cause motion sickness, eye strain and accidents if you can't see your surroundings.

DO NOW

What could go wrong if you wore a VR headset in a busy classroom?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the health, safety, privacy and ethical issues of XR and how to use it responsibly

SUCCESS CRITERIA: I CAN...

- ✓ I can describe physical health and safety risks of XR
- ✓ I can explain the privacy risks of XR data
- ✓ I can describe how to stay safe in social VR

PHYSICAL SAFETY

- You can't see real obstacles: clear the space, set a boundary
- Motion sickness, eye strain and headaches: take regular breaks
- Most headsets are not designed for young children; check the age guidance
- Keep the lenses out of direct sunlight (they can focus it and burn the screen)

KEY WORDS

GUARDIAN BOUNDARY	A virtual wall in a headset that warns you when you're near real obstacles.
BIOMETRIC DATA	Data about a person's body or behaviour, such as eye movements or fingerprints.
SOCIAL VR	Virtual spaces where people meet and interact as avatars.
AGE RATING	Guidance about the minimum age an app, game or device is suitable for.

YOUR DATA

- Headsets scan your room with cameras
- Eye tracking shows what you look at and for how long
- Body movements can be unique enough to identify you
- This data could reveal health, emotions and interests

SOCIAL VR

- Avatars meet and talk in shared virtual spaces
- Harassment can feel very real because of presence
- Use personal boundary settings, mute, block and report
- Check age ratings and never share personal information

ACTIVITY · 25 MINUTES · PAIRS

XR SAFETY GUIDE

1. Your school has bought VR headsets for lessons. Write a safety and privacy guide for Year 8 pupils.
2. Include a physical safety checklist, a "feeling unwell" plan and privacy settings to check.
3. Add a social VR code of conduct.
4. Design it as a one-page poster.

In your workbook, answer:

? Write your physical safety checklist.

? Write three rules for your social VR code of conduct.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is a guardian boundary?

A A password
B A security guard
C A firewall
D A virtual wall that warns you when you're near real obstacles

2. How often should you take a break when using VR?

A Never
B Every 20-30 minutes
C Once a week
D Every 5 hours

3. Why is eye-tracking data sensitive?

A It can reveal interests, emotions and possibly health
B It isn't
C It's very small
D It's always public

4. What should you do if someone harasses you in social VR?

A Mute, block, report and tell a trusted adult
B Ignore all safety settings
C Share your address
D Follow them

QUICK CHECK: ANSWERS

1. **D: A virtual wall that warns you when you're near real obstacles**
Keeps you safe in your space.

2. **B: Every 20-30 minutes**
Regular breaks reduce strain.

3. **A: It can reveal interests, emotions and possibly health**
Biometric data.

4. **A: Mute, block, report and tell a trusted adult**
Use the tools.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three physical health and safety risks of using a VR headset.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe physical health and safety risks of XR
- ✓ I can explain the privacy risks of XR data
- ✓ I can describe how to stay safe in social VR

LESSON 6 OF 6

DESIGN AN XR EXPERIENCE

Good XR design starts with a clear purpose and real users.

STARTER · 5 MINUTES

DO NOW

What's the best VR or AR experience you've tried or seen? What made it good?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Design a comfortable, accessible and purposeful XR experience using user-centred design and storyboarding

SUCCESS CRITERIA: I CAN...

- ✓ I can plan an XR experience for a real purpose and audience
- ✓ I can choose interaction and movement methods that are comfortable and accessible
- ✓ I can storyboard and prototype an XR experience

PURPOSE FIRST

- Does it need to be XR? What does XR add?
- Who are the users? How old? Any accessibility needs?
- What should users learn, feel or do?
- How long will they spend in it?

KEY WORDS

STORYBOARD	A series of sketches showing each scene of an experience.
INTERACTION DESIGN	Designing how users control and respond to a product.
ACCESSIBILITY	Designing so people with different abilities can use something.
PROTOTYPE	An early version used to test ideas.
USER TESTING	Watching real users try a product to find problems and improve it.

INTERACTION AND COMFORT

- Interaction: controllers, hand tracking, gaze, voice
- Movement: teleport, snap turn, seated or standing
- Keep text large and near the centre of view
- Audio cues help users know where to look

STORYBOARD, PROTOTYPE, TEST

- Storyboard each scene: what the user sees, hears and does
- Prototype with paper, or tools like CoSpaces Edu
- Test with real users; watch where they get stuck
- Improve and test again

ACTIVITY · 35 MINUTES · GROUPS OF 3

XR DESIGN SPRINT

1. Choose a brief: an AR tour of Enniskillen Castle, a VR experience showing what's inside a computer, or an AR guide for new pupils.

2. Write a purpose statement and describe your users.

3. Storyboard at least four scenes, showing what the user sees, hears and does, plus how they move and interact.

4. Add comfort and accessibility features, then test your storyboard with another group and improve it.

In your workbook, answer:

? Write your purpose statement and explain why XR suits it.

? Describe two comfort or accessibility features and one change you made after testing.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What should be the first question when designing an XR experience?

A How many levels?
B Does this need to be XR, and what does XR add?
C What colour is the logo?
D Which headset is cheapest?

2. What is a storyboard?

A A type of headset
B A wooden board
C A series of sketches showing each scene
D A book review

3. Which feature improves accessibility?

A No subtitles
B Only standing play
C Tiny text at the edge of view
D A seated mode

4. Why is user testing important?

A It shows where real users get stuck or uncomfortable
B It's not
C It makes the app longer
D It replaces design

QUICK CHECK: ANSWERS

1. **B: Does this need to be XR, and what does XR add?**
   Purpose first.
2. **C: A series of sketches showing each scene**
   Plans the experience.
3. **D: A seated mode**
   Include more users.
4. **A: It shows where real users get stuck or uncomfortable**
   Improve with evidence.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three guidelines for making a VR experience comfortable.

Look back at today's success criteria: can you tick them all?

- ✓ I can plan an XR experience for a real purpose and audience
- ✓ I can choose interaction and movement methods that are comfortable and accessible
- ✓ I can storyboard and prototype an XR experience

END OF TOPIC

VR, AR AND MIXED REALITY

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.